

SHAMIR KHAN

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Education

Bachelor of Science in Computer Science

April 2028

Wilfrid Laurier University

Waterloo, ON

- **Core CS:** Data Structures, OOP, Discrete Math, Algorithms, Computer Architecture, Digital Electronics

Experience

Full-Stack Developer | Team Lead

April 2026

ConHacks

Waterloo, ON

- **Led a team of 4 developers** to architect **InstantService**, an AI-driven PWA that **reduced client service-acquisition time by 70%** by automating natural language request classification and dispatch.
- **Engineered a multi-weighted scoring algorithm** using **FastAPI** and **Snowflake** to optimize contractor matching, resulting in a **40% increase in job volume** and **instant, under 3-min service confirmations**.
- **Architected the backend infrastructure** for tier-based pricing and verification, ensuring a high-trust environment that validates contractors via **license/insurance certificates**.

Logistics

Sept 2021 – Sept 2022

Hack the North

Waterloo, ON

- Facilitated a seamless **36-hour** event experience for **1,000+** global students by managing on-site logistics and wayfinding operations for **Canada's largest hackathon**.
- Orchestrated on-schedule registration and technical demos to support **827** attendees across **222** projects.
- Ensured zero major schedule disruptions by streamlining equipment layout and resolving bottlenecks rapidly for operations backed by **major corporate sponsors**.

Projects

InstantService | AI Emergency Dispatch | GitHubPWA | Next.js | FastAPI | Snowflake | Gemini 1.5 Pro | ElevenLabs

- Architected a mobile-first **Progressive Web App (PWA)** and **AI dispatch engine** using **Gemini 1.5 Pro** that reduced service-acquisition time by **70%** via automated request classification.
- Engineered a high-trust verification system that validates contractors via **license/insurance certificates** and performance metrics, ensuring reliable dispatch across three tiered service levels.
- Orchestrated a low-latency feedback loop using **ElevenLabs Neural Speech** to provide users with immediate AI-generated verbal confirmations synced with a **Snowflake** data layer.

Subscription Management System | GitHub

React | Node.js | Express | MongoDB | Arcjet | Upstash

- Built a full-stack subscription tracker with a **React** frontend bridging a modular **Node.js/Express MVC API** and **MongoDB**, implementing **JWT-based auth** and **Zod** schemas.
- Maximized application uptime and mitigated malicious traffic by deploying a production-ready API to **Render** with **Arcjet** security integration and production-ready rate limiting.
- Executed scheduled trial expirations and lifecycle notifications with high reliability by implementing automated event-driven serverless functions via **Upstash Workflow** and **Nodemailer**.

MSA Website (WLU MSA) | Live Site

Next.js | TypeScript | Payload CMS

- Improved navigation flows and content accessibility for a **250+ member community** by consulting on UI/UX product design and architecture for an official **React/Next.js** website.
- Enhanced core community features by optimizing validation logic and content structure for live prayer timings, halal directories, and interactive event registration forms.
- Expanded digital reach and engagement by supporting digital communications for a university organization, directly amplifying initiatives that generated **\$50K+** in charitable fundraising.

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks & Tools: React.js, Next.js, PWA, FastAPI, Tailwind CSS, Docker, Git, GitHub, DigitalOcean, Arcjet, Upstash

AI & Data: Gemini 1.5 Pro, ElevenLabs, Snowflake, MongoDB, Generative AI, Data Warehousing

Concepts: System Architecture, REST APIs, OOP, Data Structures, Algorithms, Automated Workflows